

QuantLite: 4-bit Weights for Small Budgets

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FICTIONAL TEST DOCUMENT - generated for the workshop; not a real paper.

Abstract

QuantLite quantizes the weights of a 7B-parameter language model to 4 bits. Memory use drops from 14 GB to 4 GB, and accuracy on our reasoning benchmark falls by 1.8 points.

1 Method

Weights are grouped in blocks of 128 and each block gets its own scale. Activations stay in 16-bit precision.

2 Results

Average accuracy fell from 63.5% to 61.7% (a drop of 1.8 points). Throughput rose 1.6x on a single consumer GPU.

3 Limitations

Long-context tasks lost 4.2 points. We evaluated a single model family.